Introductory Econometrics with Recitation — Quiz 2

ECON2023

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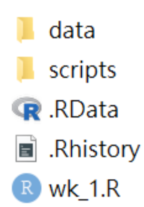
Name: ________________ **ID:** ________________

1. (**2 points**) Alice creates an R script named `wk_1.R` in her project folder. However, while working, she accidentally moves her RStudio project file, `R_code.Rproj`, into the `data` folder. The folder structure after the accident is shown in the picture below. Suppose Alice now opens the RStudio project `R_code` and then opens the script `wk_1.R`. She wants to import the file `county.csv` from the `data` folder. Which of the following lines of R code will do the job?



   A. **`county <- read.csv("county.csv")`**

   B. `county <- read.csv("data/county.csv")`

   C. **`county <- read.csv("../data/county.csv")`**

   D. `county <- read.csv("~/data/county.csv")`

2. (**2 points**) Which of the following statements is FALSE?

   A. `setwd("C:\\your\\working\\directory")` does not work on Mac.

   B. We can use `df[["col"]] <- c(1, 2)` to add a new column to `df`.

   C. `t.test(df$weight~df$habit)` performs an independent two-sample t-test.

   D. The main reason to use `install.packages()` only when needed is to prevent function name conflicts.

3. (**2 points**) What is the output of the following code?

```
df <- data.frame(a = c(2, 3, 4), b = c(4, 5, 6))
df$r1 <- df %>% rowSums()
df$r2 <- df %>% rowSums()
df %>% colSums()
```

   A.

```
 9 15 24 24
```

B.

```
a  b r1 r2
9 15 24 24
```

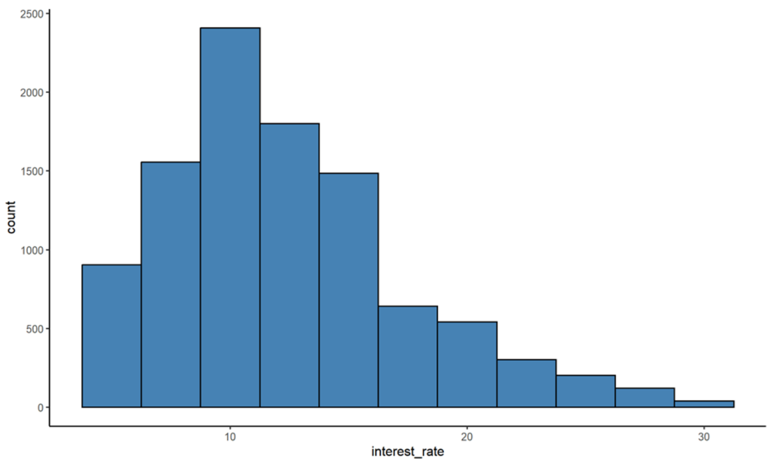
C.

```
9 15 24 48
```

D.

```
a  b r1 r2
9 15 24 48
```

4. (**2 points**) Suppose Grimm wants to plot the distribution of values in the `interest_rate` variable of the `loan` data frame. Which of the following code snippets would most likely produce the graph below?



A.

```
ggplot(loan, aes(x = interest_rate)) +
  geom_histogram(binwidth = 2.5, fill = "steelblue")
```

B.

```
ggplot(loan, aes(x = interest_rate, y = count)) +
  geom_histogram(binwidth = 2.5, fill = "steelblue")
```

C.

```
ggplot(loan, aes(x = "interest_rate")) +
  geom_histogram(binwidth = 2.5, fill = "steelblue")
```

D.

```
ggplot(loan, aes(x = "interest_rate", y = "count")) +
  geom_histogram(binwidth = 2.5, fill = "steelblue")
```

5. (**2 points**) What is the output of the following code?

```
sales <- data.frame(
  region = c("N","N","N","N","S","S","S","S"),
  year   = c(2025,2025,2025,2026,2025,2026,2026,2026),
  rep    = c("A","A","B","B","A","A","B","B"),
  amount = c(50, 80, 90, 120, 110, 120, 50, 40),
  cost   = c(40, 70, 60, 100, 90, 110, 60, 30)
)
stores <- sales %>%
  group_by(region, year, rep) %>%
  mutate(growth = amount - cost) %>%
  summarize(mean_growth = mean(growth)) %>%
  filter(mean_growth == max(mean_growth))

sum(stores$mean_growth)
```

A. `30`

B. `50`

C. `80`

D. `90`

Below is the (slightly modified) context and answer code for Week 3's practices. Please read the content and answer questions 6-10:

Context: Yuri is trapped in a time loop! In each loop, 15 possible people may appear. After repeating the loop many times, he begins to suspect that Yuriko appears more often than the other people. To investigate this, Yuri decides to record which people appear in each of the next 50 loops, so the dataset contains 50 rows. The dataset contains:

- `loop`: The loop number.
- `Yuri`, `Setsu`, `Gina`, `SQ`, `Raqio`, `Stella`, `Shigemichi`, `Chipie`, `Comet`, `Jonas`, `Kukrushka`, `Otome`, `Sha-Ming`, `Remnan`, and `Yuriko`: Each of these columns records whether the person appeared in that loop, using `1` for "appeared" and `0` for "did not appear".

In each recorded loop, Setsu and Yuriko always appear, and at least five people appear in each loop. Based on Yuri's observations, there are about 10 people in each loop on average. Because Setsu is also a time looper, she appears in every loop as well. To account for this, Yuri removes Setsu's data when calculating the appearance rate.

Yuri performs a paired-sample t-test with the following hypotheses:

$$H_0 : \text{Yuriko has the same appearance rate as the other people.}$$

$$H_1 : \text{Yuriko has a different appearance rate from the other people.}$$

Yuri wants to know: after how many loops can he reject the null hypothesis at $p < 0.0001$ and conclude that Yuriko does not have the same appearance rate as the other people?

Answer Code:

```r
library(tidyverse)

gnosia_raw <- read.csv("data/gnosia_raw.csv")
gnosia_tidy <- gnosia_raw %>% select(-c(loop, Yuri, Setsu))

t_test_pval <- function(data){
  ## create column appear_sum that calculates the row sum of all columns
        (Code 1)
  ## calculate non-Yuriko appearance rate
        (Code 2)
  ## conduct a paired sample t-test
  test <- t.test(data$Yuriko, data$not_Yuriko_appear_rate, paired = TRUE)
  ## extract the p-value from the t-test
        (Code 3)
  return(p)
}

minimal_loop <- 0
finish_index <- 0

for(i in 2:50){
  current_loop_data <- gnosia_tidy[1:i, ]
  pval <- t_test_pval(current_loop_data)

  if(pval < 0.0001 && finish_index == 0){
    minimal_loop <- i
    finish_index <- 1
  }
  print(pval)
}
print(minimal_loop)
```

6. (**2 points**) Which of the following statements is FALSE?

   A. The following code works for (Code 1).

```r
data$appear_sum <- data %>% rowSums()
```

   B. The following code works for (Code 1).

```r
data <- data %>% mutate(appear_sum = rowSums())
```

   C. The following code works for (Code 2).

```r
data$not_Yuriko_appear_rate <- (data$appear_sum - data$Yuriko) / 12
```

   D. The following code works for (Code 2).

```r
data<-data %>% mutate(not_Yuriko_appear_rate=(appear_sum-Yuriko)/12)
```

7. (**2 points**) Which of the following code does NOT work for (Code 3)?

   A.
   ```
   p <- test$p.value
   ```

   B.
   ```
   p <- test$"p.value"
   ```

   C.
   ```
   p <- test[[p.value]]
   ```

   D.
   ```
   p <- test[["p.value"]]
   ```

8. (**2 points**) Which of the following statements is FALSE?

   A. The code still works if we replace `library(tidyverse)` with `library(dplyr)` in Line 1.

   B. The code still works if we replace `gnosia_tidy[1:i, ]` with `gnosia_tidy[1:i]` in Line 22.

   C. The code still works if we replace `pval < 0.0001 && finish_index == 0` with `pval < 0.0001 & finish_index == 0` in Line 25.

   D. Creating a new function `t_test_pval` is not necessary for the code to work. That is, there is an equivalent way to write the code without using `function()` and still obtain the same result as the original code.

9. (**2 points**) Which of the following statements is TRUE?

   A. The code includes `finish_index` inside the `for` loop to guarantee that `minimal_loop` is always assigned a value.

   B. The function `t_test_pval` can be saved in a script and called using `load("t_test_pval.R")`.

   C. Since Yuriko always appears, and the appearance rate of others ≤ 1 in each loop, the printed `pval` values should form a weakly decreasing sequence.

   D. Even though there is no full appearance in every loop (≈ 10 people in each loop on average), and we skip the first loop in Line 21, the code may still fail.

10. (**2 points**) Consider the following three ways to calculate the p-value in `t_test_pval()`:

```
test_1 <- t.test(data$not_Yuriko_appear_rate, mu = 1)
p_1 <- test_1$p.value

test_2 <- t.test(data$not_Yuriko_appear_rate, data$Yuriko, paired = TRUE)
```

```
p_2 <- test_2$p.value

test_3 <- t.test(data$not_Yuriko_appear_rate, data$Yuriko)
p_3 <- test_3$p.value
```

Which of the following statements best describes the relationship among p_1, p_2, and p_3?

- **A.** $p_1 = p_2 = p_3$
- B. $p_1 < p_2 = p_3$
- C. $p_1 < p_2 < p_3$
- D. $p_1 < p_3 < p_2$

Below is the context and (simplified) answer code for Week 4's practices. Please read the content and answer questions 11-15:

Context: Inori is practicing for the Chubu Regional Championships, a figure skating competition in Japan. Her coach, Tsukasa, records the result of each jump attempt every day during practice. The dataset contains:

- `date`: The date of the practice session.
- `jump`: The type of figure skating jump.
- `result`: The landing outcome of the jump attempt.

On March 23, Inori caught a mild cold, and Tsukasa noticed that her landing performance seemed to be affected. Help Tsukasa test his hypothesis.

Answer code:

```
library(tidyverse)

cold_date <- as.Date("2026-03-23")
medalist_raw <- read_csv("medalist.csv")

medalist_tidy <- medalist_raw %>%
  mutate(date = as.Date(date)) %>%
  mutate(is_landing = if_else(result == "Landing", 1, 0)) %>%
  mutate(result = factor(result,levels=c("Fall","Touch down","Landing"))) %>%
  mutate(is_after_cold=if_else(date<cold_date,"before_cold","after_cold")) %>%
  mutate(is_after_cold=factor(is_after_cold,levels=c("before_cold","after_cold")))

medalist_bar_plot <- medalist_tidy %>%
  ggplot(aes(x = result, fill = is_after_cold)) +
  geom_bar(position = "dodge") +
  labs(title = "Jump results", x = "Result", y = "Count")

medalist_line_plot <- medalist_tidy %>%
  ggplot(aes(x = date, y = is_landing)) +
```

```
20   geom_line(stat = "summary",
21             fun = "mean",
22             linewidth = 0.7,
23             color = "steelblue") +
24   geom_vline(xintercept = cold_date, color = "lightcoral") +
25   geom_hline(yintercept = 0.8, color = "lightcoral") +
26   labs(title = "Daily average landing success rate",
27        x = "Date", y = "Success rate")
28
29 medalist_success_rate_by_date_jump <- medalist_tidy %>%
30   group_by(date, jump) %>%
31   summarise(success_rate = mean(is_landing)) %>%
32   ungroup()
33
34 medalist_facet_line_plot <- medalist_success_rate_by_date_jump %>%
35   ggplot(aes(x = date, y = success_rate)) +
36   geom_line(linewidth = 0.7,
37             color = "steelblue") +
38   geom_vline(xintercept = cold_date, color = "lightcoral") +
39   geom_hline(yintercept = 0.8, color = "lightcoral") +
40   facet_wrap(~ jump) +
41   labs(title = "Daily landing success rate (by jump)",
42        x = "Date", y = "Success rate")
43
44 ggsave(filename = "medalist_bar_plot.png",
45        plot = medalist_bar_plot,
46        path = "output",
47        width = 12,
48        height = 8,
49        dpi = 300)
50
51 ggsave(filename = "medalist_facet_line_plot.png",
52        plot = medalist_facet_line_plot,
53        path = "output",
54        width = 12,
55        height = 8,
56        dpi = 300)
```

11. (**2 points**) What happens to the output of `medalist_line_plot` if we replace `cold_date` with `"2026-03-23"` in Line 24?

    A. It immediately returns an error.

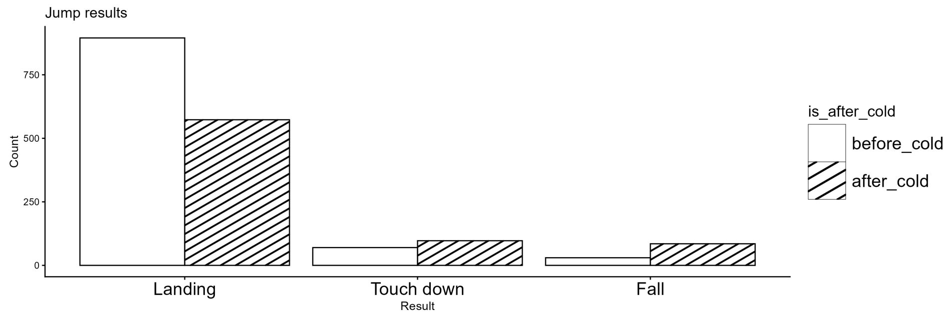
    B. `ggplot()` will try to convert the string to a date type. Since `"2026-03-23"` can be converted to a date type, the output should be the same.

    C. Because `ggplot()` cannot determine where to place the line on the x-axis, it generates a plot without any vertical lines.

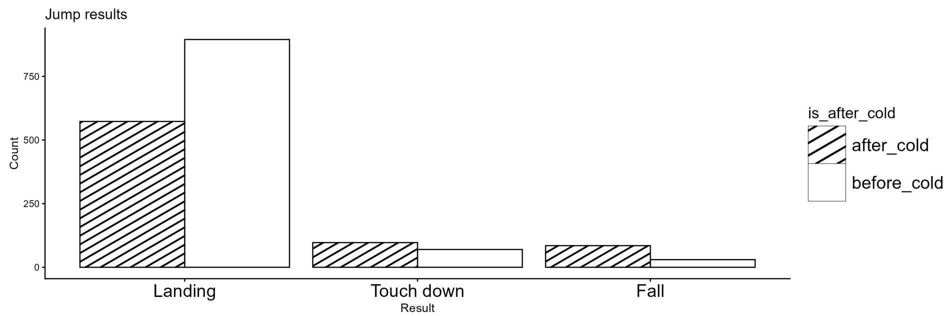
D. `ggplot()` will coerce `cold_date` to a numeric type. The plot will contain a vertical line if the coerced number lies within the range of the x-axis; otherwise, it will display a warning after the plot is shown.

12. (**2 points**) What will `medalist_bar_plot` look like if we remove Lines 9 and 11, as well as the `%>%` in Line 10?

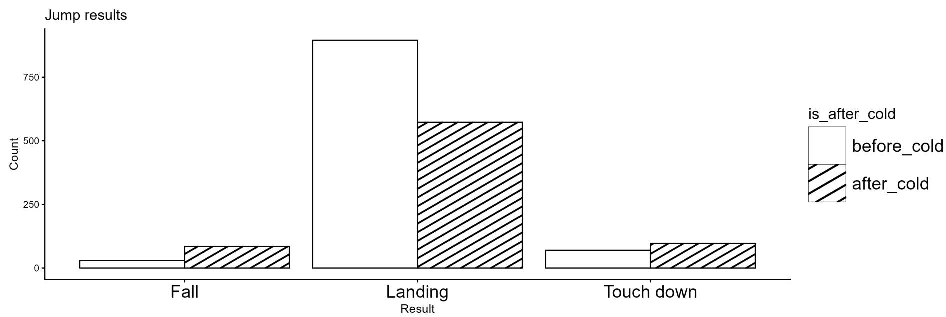
A.



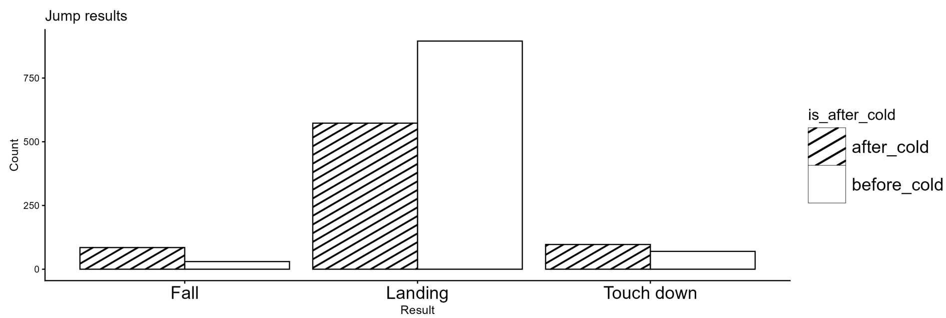
B.



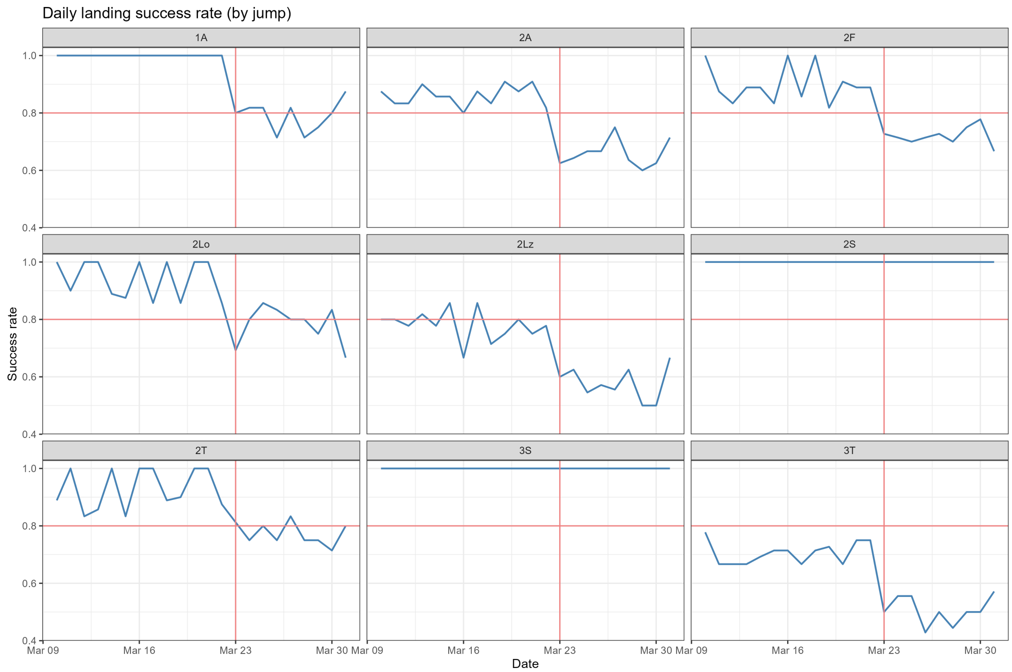
C.



D.



13. (**2 points**) The figure below shows `medalist_facet_line_plot`. Which of the following statements about this graph is FALSE?



A. Inori's jump performance was affected by the mild cold.

B. Adding vertical and horizontal lines not only makes the plot clearer, but also implicitly causes `ggplot()` to use the same range for the x- and y-axes.

C. There are some jumps that appear not to be affected by the cold.

D. The plot remains the same if we add `stat = "summary", fun = "mean",` to `geom_line()` in Line 36.

14. (**2 points**) Which of the following statements is FALSE?

A. The code below is an equivalent way to generate `medalist_line_plot`:

```
medalist_line_plot <- medalist_tidy %>%
  group_by(date) %>%
  summarise(success_rate = mean(is_landing)) %>%
  ungroup() %>%
  ggplot(aes(x = date, y = success_rate)) +
  geom_line(linewidth = 0.7, color = "steelblue") +
  geom_vline(xintercept = cold_date, color = "lightcoral") +
  geom_hline(yintercept = 0.8, color = "lightcoral") +
  labs(title = "Daily average landing success rate",
       x = "Date", y = "Success rate")
```

B. Placing `%>%` at the end of each line not only improves readability, but also prevents errors that would occur if `%>%` were moved to the beginning of the next line.

C. Forgetting to call `ungroup()` in Line 32 will cause a problem here: it affects the output of `medalist_facet_line_plot`.

D. We can merge all `mutate()` to a single one from Line 7 to 11:

```
medalist_tidy <- medalist_raw %>%
  mutate(
    date = as.Date(date),
    is_landing = if_else(result == "Landing", 1, 0),
    result = factor(result,levels=c("Fall","Touch down","Landing")),
    is_after_cold=if_else(date<cold_date,"before_cold","after_cold"),
    is_after_cold=factor(is_after_cold,levels=c("before_cold","
    ↪ after_cold")))
```

15. (**2 points**) Which of the following statements about `ggsave()` is correct?

A. We can save two plots at the same time if they have the same `path`, `width`, `height`, and `dpi` settings. That is, we can replace Lines 44 to 56 with

```
ggsave(
  filename = c("medalist_bar_plot.png", "medalist_facet_line_plot"),
  plot = c(medalist_bar_plot, medalist_facet_line_plot),
  path = "output",
  width = 12,
  height = 8,
  dpi = 300
)
```

B. In Lines 51 to 56, changing `width = 12, height = 8` to `width = 6, height = 4` not only makes the saved image smaller, but also changes the layout of the graph.

C. If the folder specified in `path = "output"` does not exist, `ggsave()` immediately returns an error and says it cannot find the directory.

D. If we do not specify the file type to save (for example, `filename = "medalist_hist_plot"`), it will be saved as a JPEG file by default.

Answer Key

1.A 2.D 3.D 4.A 5.C 6.B 7.C 8.B 9.D 10.A 11.A 12.D 13.B 14.C 15.B